

Operating Systems

CSIT — 4th Semester — 2081 — Answer Sheet
Subject Code: CSC264

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Explain the translation of logical address into physical address using segment table with necessary diagram. List advantages and disadvantages of segmentation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Find the seek time using SCAN, C-SCAN, Look and C-Look disk scheduling algorithm for processing the following requests queue: 35, 70, 45, 15, 65, 20, 80, 90, 75, 130. Suppose the disk has tracks numbered from 0 to 150 and assume the disk arm to be at 30 and moving outward.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain Sleeping Barber problem. Illustrate on how it can be solved.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Explain microkernels and exokernels.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Consider a swapping system in which memory consists of the following hole sizes in memory order 15 MB, 2 MB, 10 MB, 6 MB, 8 MB and 20 MB. Which hole is taken for successive segment requests of a) 6 MB b) 10 MB c) 8 MB for first fit, next fit and best fit.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain how semaphore solves the problem of critical section?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. How do you think deadlock can be avoided? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Explain Inter-Process Communication in Linux.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. List different file structures and explain them.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Calculate the average waiting time and turn around time using priority algorithm (priority 1 being the highest) for the given scenario:

PID	Burst Time (s)	Arrival Time	Priority
A	40	3	2
B	22	3	3
C	42	2	1
D	23	1	1

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Explain memory mapped IO.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on a) Virtual memory b) Race condition

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.